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Transformers, Time Series, and the Myth of Permutation Invariance

Beware the false prophets of forecasting!

NIKOS KAFRITSAS

OCT 17



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Validation error

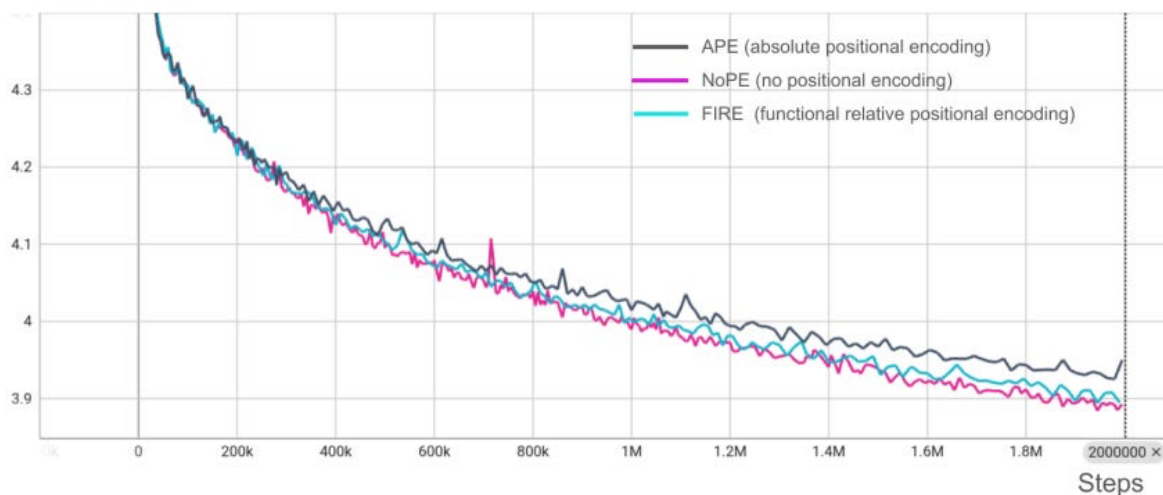


Figure 1. NoPE(no positional embeddings) outperforms APE (absolute positional encodings) on TimesFM-ICF paper
(Source [1])

After many years of blogging about time series, **one myth just won't die.**

That Transformers shouldn't be used for forecasting because attention is permutation-invariant.

This is misused. Since 2020, nearly all major Transformer forecasting models encode order through other means or redefine attention itself.

Google's [TimesFM-ICF\[1\]](#) paper confirms what we knew: Their experiments show the model performs just as well with or without positional embeddings. (**Figure 1**)

Sadly, the myth will live on—kept alive by influential experts who sell books and courses to thousands. If you're new, remember: Forecasting Transformers are just great tools, not miracles or mistakes.

In this article, we'll take a closer look at this myth and examine what's true and what isn't.

✓ Find a tutorial notebook on TimesFM, along with other cool projects here: [AI Projects Folder \(Project 24\)](#)

Preliminaries

Let's start with the basics — what is attention?

Given a sentence, attention calculates the relative importance of each word compared to every other word, as pairwise scores.

The attention mechanism isn't new. It's mathematically equivalent to kernel regression (the Exponential Dot Product), but without gradients [Nadaraya-Watson, 1964]. Let's see how it works:

Consider the sentence "**music inspires emotion**". **Figure 2**(top) shows the full-attention matrix weights, where $a_{i,j}$ is the attention between words

at positions i and j .

Now, let's permute the words to create the sentence "**emotion inspires music**". Since attention is calculated as pairwise scores between words, we can also permute the respective attention weights (**Figure 2** bottom):

	emotion	inspires	music
emotion	$a_{1,1}$	$a_{1,2}$	$a_{1,3}$
inspires	$a_{2,1}$	$a_{2,2}$	$a_{2,3}$
music	$a_{3,1}$	$a_{3,2}$	$a_{3,3}$

	music	inspires	emotion
music	$a_{3,3}$	$a_{3,2}$	$a_{3,1}$
inspires	$a_{2,3}$	$a_{2,2}$	$a_{2,1}$
emotion	$a_{1,3}$	$a_{1,2}$	$a_{1,1}$

Figure 2: Full-attention matrices for the sentences "emotion inspires music" and "music inspires emotion" (image by author)

Therefore, while the 2 sentences have different meanings, their permutation leads to the same attention scores — hence attention is *permutation invariant*.

That's why we add extra components to encode position. The original Transformer used absolute position encodings.

Back to time-series. Since attention is permutation-invariant, and order is crucial in time-series (even more than in NLP), why use it at all?

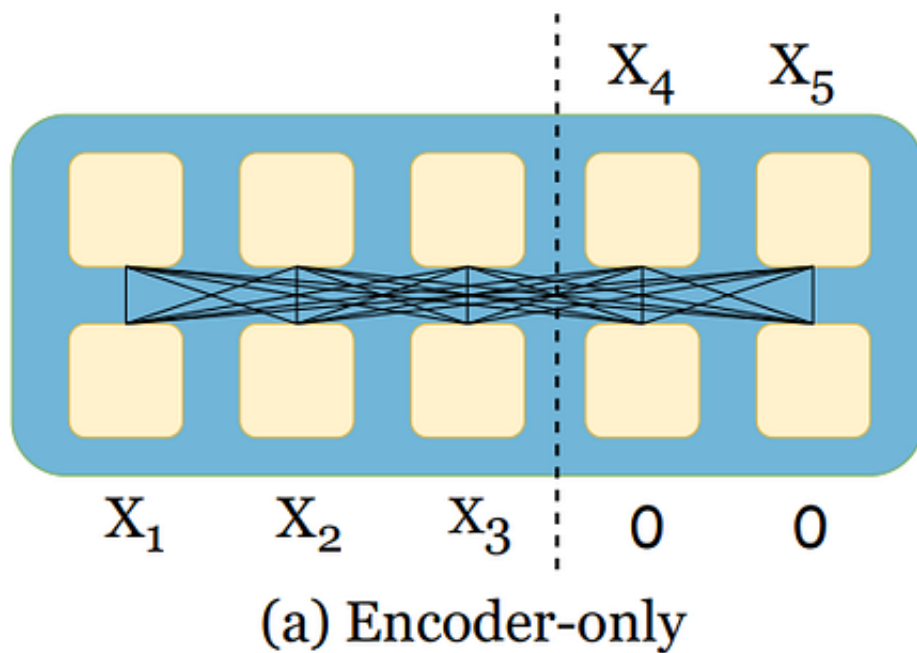
The answer is simple: there are different types of attention — and they behave differently.

Enter Causal Attention

Figure 2 illustrates a particular type of attention, known as quadratic self-attention.

This form is used by masked, encoder-only models (**Figure 3**, top). But with the rise of LLMs, attention has evolved fast.

The key idea behind the newer variants is causal attention, used by modern decoder-only models (LLMs) (**Figure 3**, bottom).



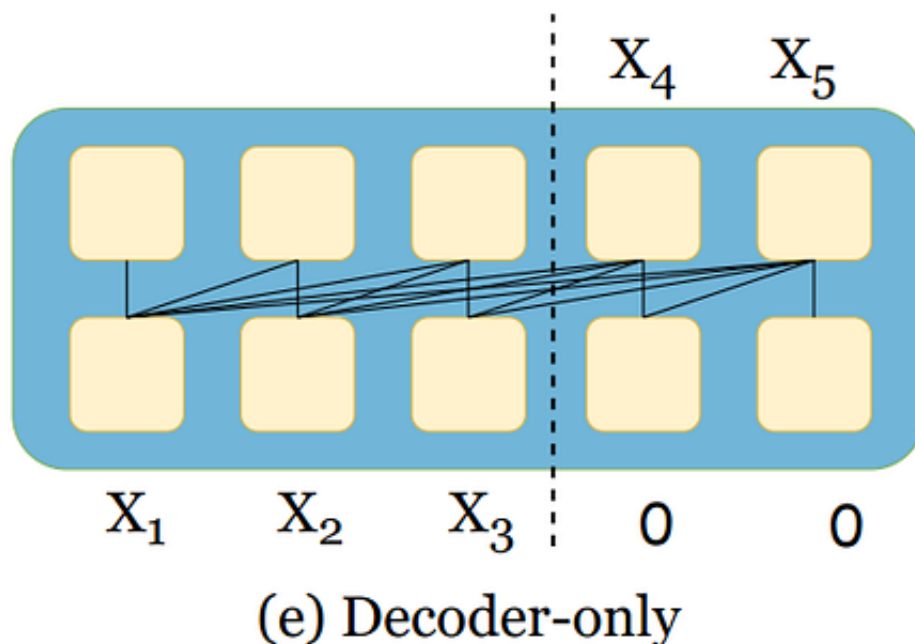


Figure 3: *Encoder-only vs Decoder-only architectures*
(Source [2])

- **Self-attention models** use bidirectional attention (looking at past and future context). They excel at tasks like imputation and anomaly detection.
- **Causal-attention models** use unidirectional attention (looking only backward) to predict the next value in a sequence, making them suitable for generation (forecasting).

However, recent research has shown that each type of attention has different properties, one of which is related to positional embeddings.

We'll discuss this next.

Enter TimesFM-ICF

Google's **TimesFM-ICF** paper revealed something remarkable—yet counterintuitive (at least to those new to this space).

TimesFM-ICF, a decoder-only foundation model using causal attention, performs *just as well* with or without positional embeddings.

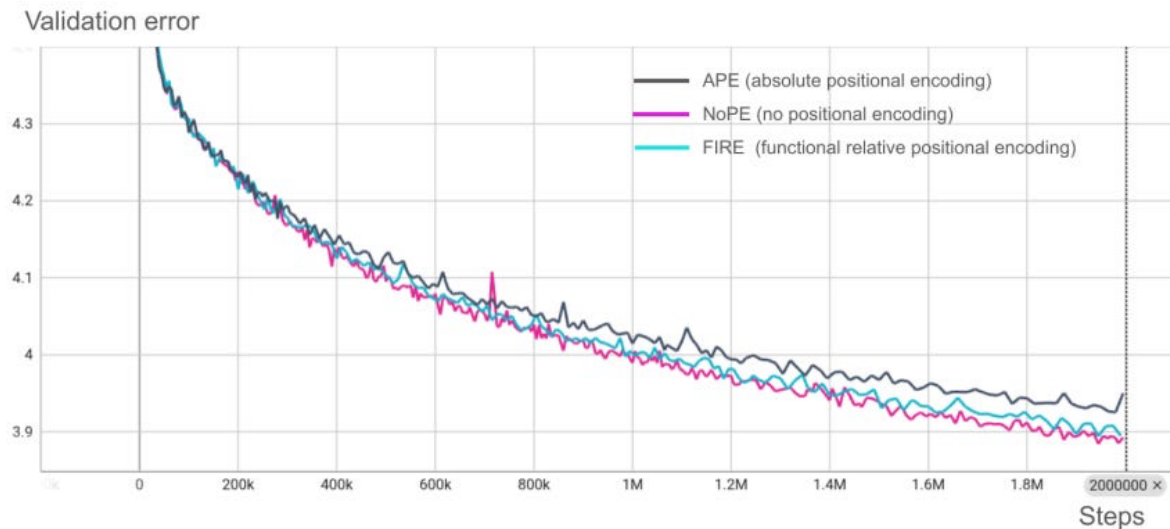


Figure 4. *NoPE(no positional embeddings) outperforms APE (absolute positional encodings) (Source [1])*

This happens for 2 main reasons:

1. **Causal attention** (unlike standard self-attention) implicitly encodes positional information once a single Transformer layer is present. This effect compounds in a well-pretrained, generalizable model.
2. **MLPs within each Transformer block** inherently capture positional patterns. That's why linear DL models like TSMixer are also successful.

Let's discuss both reasons in detail

The first reason comes from another paper [3], which TimesFM-ICF references. In it, the authors test how positional embeddings affect GPT-2—also a decoder-only model.

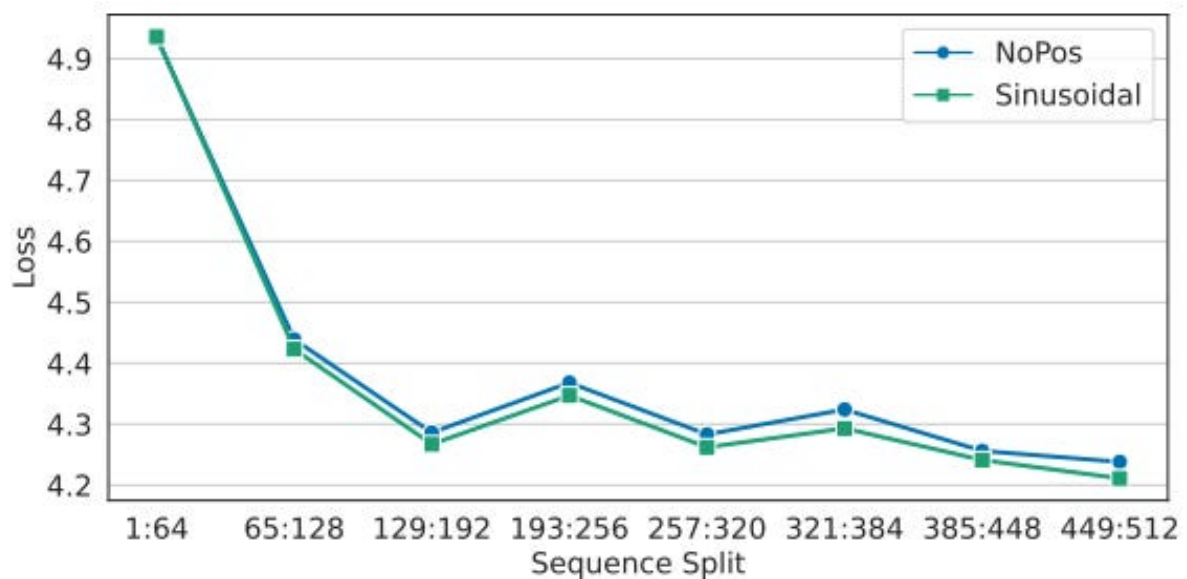


Figure 5: *The NoPos model performs comparably to the baseline with learned absolute positional encoding across all parts of the sequence. (Source [3])*

Although the figures differ slightly, there's a small but noticeable gap in GPT-2 performance with and without position embeddings. However, in forecasting models, another factor comes into play.

From time series DL papers, we know MLPs also help capture positional information. The TSMixer authors highlight this: *"The time-step-dependent linear model, despite its simplicity, proves to be highly effective in modeling temporal patterns."* [4]

That's why forecasting models often add extra MLP layers (more than NLP Transformers of similar size) to better model temporal dynamics. TimesFM, for example, includes additional input and output embedding MLPs.

Therefore, adding extra MLPs bridges the performance gap between **Figure 4** and **Figure 5** (the small difference with or without embeddings).

Fortunately, **NoPE** setup brings key advantages during pretraining:

- **Better length generalization** — crucial when extending context windows with in-context examples, as NoPE models handle longer prompts more gracefully.
- **Semantic consistency** — removing absolute encodings avoids a mismatch between base model training (without examples) and continued pretraining (with added context), preventing positional drift.
- **No accuracy trade-off** — empirically, NoPE maintains validation performance comparable to advanced length-generalizing encodings like FIRE.

Therefore, we can conclude:

Evidence shows that decoder-only, pretrained time-series models can perform equally well without positional embeddings. Attention alone drives performance, enabling flexible context and prediction lengths.

The Pre-2024 Era

Decoder-based Transformers for forecasting are mostly a recent trend—but what came before?

Most early Transformer forecasting models were **encoder-based**, encoding position through alternative mechanisms. For example:

- **Hybrid architectures:** Models like TFT used an LSTM encoder-decoder to capture local dependencies.
- **Feature-based attention:** Models like iTransformer applied attention

across features (variates), not time.

I've discussed these architectures in more detail here:



Will Transformers Revolutionize Time-Series Forecasting? Advanced Insights, Part 1

NIKOS KAFRITSAS · JUNE 25, 2024

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Will Transformers Revolutionize Time-Series Forecasting? - Advanced Insights, Part 2

NIKOS KAFRITSAS · JULY 27, 2024

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Many of these earlier models showed limited improvement over simpler deep learning approaches. But that wasn't due to attention's permutation invariance—it was **overfitting**, as the TSMixer paper explains [4].

effective in modeling temporal patterns. Conversely, even though recurrent or attention architectures have high representational capacity, achieving time-step independence is challenging for them. They usually overfit on the data instead of solely considering the positions. This unique property of linear models may help explain the results in Zeng et al. (2023), where no other method was shown to match the performance of the linear model.

Source [4]

Put simply, these models were trained on toy datasets—too small for a Transformer's capacity.

By contrast, **foundation time-series models**, trained on trillions of tokens, do leverage attention effectively. When that attention is **causal**, performance soars—often surpassing even supervised models (**Figure 6**).

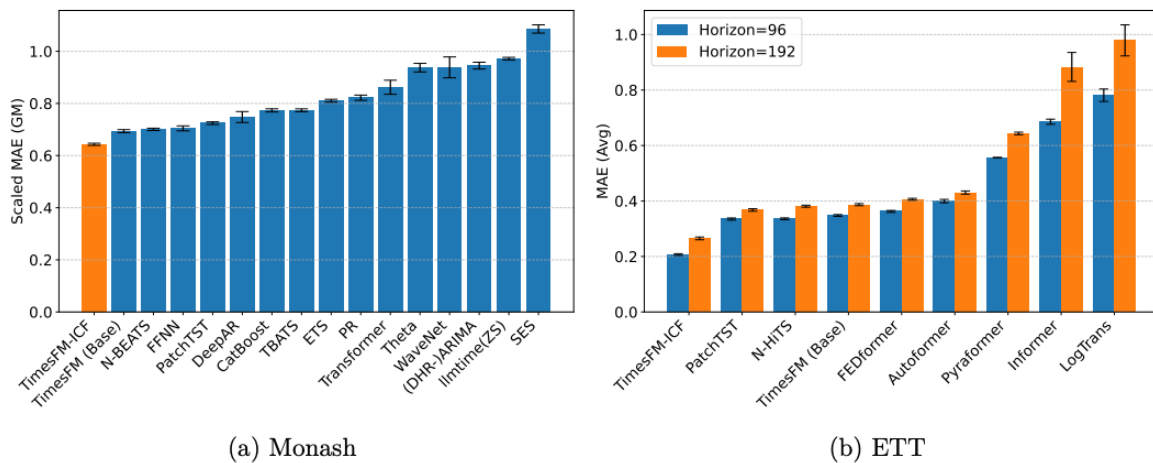


Figure 6: Performance results: (a) shows the geometric mean of scaled MAE on Monash datasets, where TimesFM (base) outperforms the next best baseline by 7%. (b) reports average MAE on four ETT datasets using rolling validation, with TimesFM-ICF achieving at least a 25% improvement over all baselines. (Source [1])

I'll skip the deeper details here. You can find more in my previous article on [TimesFM-2.5](#) and [TimesFM-ICF](#).

Ignoring the Noise

Until proven otherwise, **causal attention** remains superior to self-attention for time-series forecasting at the current scales we study these models.

Yet, some papers attempt to claim otherwise. One example, shown in [5], appears at first glance to contradict the findings of **TimesFM-ICF**, suggesting serious limitations in the model:

Why Do Transformers Fail to Forecast Time Series In-Context?

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Abstract

Time series forecasting (TSF) remains a challenging and largely unsolved problem in machine learning, despite significant recent efforts leveraging Large Language Models (LLMs), which predominantly rely on Transformer architectures. Empirical evidence consistently shows that even powerful Transformers often fail to outperform much simpler models, e.g., linear models, on TSF tasks; however, a rigorous theoretical understanding of this phenomenon remains limited. In this paper, we provide a theoretical analysis of Transformers' limitations for TSF through the lens of In-Context Learning (ICL) theory. Specifically, under $AR(p)$ data, we establish that: (1) Linear Self-Attention (LSA) models *cannot* achieve lower expected MSE than classical linear models for in-context forecasting; (2) as the context length approaches to infinity, LSA asymptotically recovers the optimal linear predictor; and (3) under Chain-of-Thought (CoT) style inference, predictions collapse to the mean exponentially. We empirically validate these findings through carefully designed experiments¹. Our theory not only sheds light on several previously underexplored phenomena but also offers practical insights for designing more effective forecasting architectures. We hope our work encourages the broader research community to revisit the fundamental theoretical limitations of TSF and to critically evaluate the direct application of increasingly sophisticated architectures without deeper scrutiny.

Source [5]

Here's the catch, though: This paper studies self-attention, and specifically linear self-attention.

While the authors present valuable theoretical insights, their findings are largely irrelevant to the current state of the art. Moreover, they fail to cite the TimesFM-ICF preprint, published nearly a year earlier.

Closing Remarks

Normally, I wouldn't bother publishing an article like this — these distinctions should be obvious to any impartial, well-informed practitioner.

My goal is to bring clarity and help you focus on what truly matters.

After all, Transformers aren't a silver bullet; they're just tools — powerful

ones, but only when used in the right context!

Thank you for reading!

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References

1. Das et al. [*In-context Fine-tuning for Time-series Foundation Models \(ICML 2025\)*](#)
2. Shen et al. [*The Power of Architecture: Deep Dive into Transformer Architectures for Long-Term Time Series Forecasting*](#)
3. Havivt et al. [*Transformer Language Models without Positional Encodings Still Learn Positional Information*](#)
4. Chen et al. [*TSMixer: An All-MLP Architecture for Time Series Forecasting*](#)
5. Zhou et al. [*Why Do Transformers Fail to Forecast Time Series In-Context?*](#)

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